LAB SHEET 3

OBJECTIVE

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

DDL Questions (Data Definition Language)

1. Create the **TEACHER** table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

   QUERY

```sql
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
CREATE TABLE TEACHER (
    Name VARCHAR(30),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(30),
    Location VARCHAR(30),
    Salary DECIMAL(10,2),
    PRIMARY KEY (DeptNo)
);
DESC TEACHER;
```

   OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell

Field	Type	Null	Key	Default	Extra
Name	varchar(30)	YES		NULL	
DeptNo	int(11)	NO	PRI	NULL	
Date_of_Joining	date	YES		NULL	
DeptName	varchar(30)	YES		NULL	
Location	varchar(30)	YES		NULL	
Salary	decimal(10,2)	YES		NULL	

2. Create the **SALES** table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(30),
    Branch VARCHAR(30),
    Salesamount DECIMAL(10,2),
    DOB DATE
);
DESC SALES;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell

Field	Type	Null	Key	Default	Extra
Sales_No	int(11)	NO	PRI	NULL	
Salesname	varchar(30)	YES		NULL	
Branch	varchar(30)	YES		NULL	
Salesamount	decimal(10,2)	YES		NULL	
DOB	date	YES		NULL	

3. Create the **DEPARTMENTS** table with the following attributes: (dept_no, dept_name, dept_location).

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(30),
    dept_location VARCHAR(30)
);
DESC DEPARTMENTS;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

Field	Type	Null	Key	Default	Extra
dept_no	int(11)	NO	PRI	NULL	
dept_name	varchar(30)	YES		NULL	
dept_location	varchar(30)	YES		NULL	

4. Create the **EMPLOYEES** table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(30),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
DESC EMPLOYEES;
```

OUTPUT

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
emp_id	int(11)	NO	PRI	NULL	
emp_name	varchar(30)	YES		NULL	
emp_salary	decimal(10,2)	YES		NULL	
dept_no	int(11)	YES	MUL	NULL	

DML Questions (Data Manipulation Language)

From TEACHER Table

1. Insert five records into the **TEACHER** table.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
INSERT INTO TEACHER VALUES
('Anita Sharma',101,'2015-06-15','Mathematics','Delhi',60000),
('Rahul Mehta',102,'2018-08-20','Commerce','Mumbai',55000),
('Priya Verma',103,'2020-01-10','Physics','Bangalore',50000),
('Amit Kapoor',104,'2016-04-18','Mathematics','Chennai',65000),
('Neha Singh',105,'2019-09-25','Commerce','Pune',58000);

SELECT * FROM TEACHER;
```

OUTPUT

Result Grid | Filter Rows: | Edit: | Export/Import:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita Sharma	101	2015-06-15	Mathematics	Delhi	60000.00
Rahul Mehta	102	2018-08-20	Commerce	Mumbai	55000.00
Priya Verma	103	2020-01-10	Physics	Bangalore	50000.00
Amit Kapoor	104	2016-04-18	Mathematics	Chennai	65000.00
Neha Singh	105	2019-09-25	Commerce	Pune	58000.00
NULL	NULL	NULL	NULL	NULL	NULL

2. Give an increment of 25% salary to teachers working in the Mathematics department.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';

SELECT * FROM TEACHER;
```

OUTPUT

Result Grid | Filter Rows: | Edit: | Export/Import:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita Sharma	101	2015-06-15	Mathematics	Delhi	75000.00
Rahul Mehta	102	2018-08-20	Commerce	Mumbai	55000.00
Priya Verma	103	2020-01-10	Physics	Bangalore	50000.00
Amit Kapoor	104	2016-04-18	Mathematics	Chennai	81250.00
Neha Singh	105	2019-09-25	Commerce	Pune	58000.00
NULL	NULL	NULL	NULL	NULL	NULL

3. Give an increment of 15% salary to teachers working in the Commerce department.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';

SELECT * FROM TEACHER;
```

OUTPUT

Result Grid | Filter Rows: | Edit: | Export/Import:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anita Sharma	101	2015-06-15	Mathematics	Delhi	75000.00
Rahul Mehta	102	2018-08-20	Commerce	Mumbai	63250.00
Priya Verma	103	2020-01-10	Physics	Bangalore	50000.00
Amit Kapoor	104	2016-04-18	Mathematics	Chennai	81250.00
Neha Singh	105	2019-09-25	Commerce	Pune	66700.00
NULL	NULL	NULL	NULL	NULL	NULL

From SALES Table

1. Insert five records into the **SALES** table.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
INSERT INTO SALES VALUES
(201,'Rohan','Delhi',25000,'1995-12-21'),
(202,'Anita','Mumbai',30000,'1998-07-15'),
(203,'Karan','Delhi',28000,'1997-12-05'),
(204,'Meena','Bangalore',35000,'1996-03-10'),
(205,'Amit','Mumbai',40000,'1999-12-18');

SELECT * FROM SALES;
```

OUTPUT

Result Grid | Filter Rows: | Edit:

Sales_No	Salesname	Branch	Salesamount	DOB
201	Rohan	Delhi	25000.00	1995-12-21
202	Anita	Mumbai	30000.00	1998-07-15
203	Karan	Delhi	28000.00	1997-12-05
204	Meena	Bangalore	35000.00	1996-03-10
205	Amit	Mumbai	40000.00	1999-12-18
NULL	NULL	NULL	NULL	NULL

2. Calculate the total sales amount in each branch.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT

Result Grid | Filter Rows:

Branch	Total_Sales
Bangalore	35000.00
Delhi	53000.00
Mumbai	70000.00

3. Calculate the average sales amount in each branch.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT

Result Grid | Filter Rows:

Branch	Average_Sales
Bangalore	35000.000000
Delhi	26500.000000
Mumbai	35000.000000

4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT Salesname,
       DATE_FORMAT(DOB,'%d-%b-%y') AS Formatted_DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT

Result Grid | Filter Rows:

Salesname	Formatted_DOB
Rohan	21-Dec-95
Karan	05-Dec-97
Amit	18-Dec-99

5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT Salesname,
       DATE_FORMAT(DOB,'%d-%b-%y') AS Formatted_DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

OUTPUT

Result Grid | Filter Rows:

Salesname	Formatted_DOB
Rohan	21-Dec-95
Karan	05-Dec-97
Amit	18-Dec-99
Anita	15-Jul-98
Meena	10-Mar-96

From JOIN Operations

1. Display employee details and department details where the department numbers are the same in both tables.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT *
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT

2. Display the employee name and department name using a LEFT OUTER JOIN.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT

Result Grid | Filter Rows:

emp_name	dept_name
Rahul	CSE
Priya	ECE
Amit	IT
Neha	CSE
Karan	MECH
Anita	ECE

3. Display the employee name and department name using a RIGHT OUTER JOIN.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

OUTPUT

Result Grid | Filter Row

emp_name	dept_name
Rahul	CSE
Neha	CSE
Priya	ECE
Anita	ECE
Amit	IT
Karan	MECH

4. Display the details of employees who draw a salary greater than the average salary.

QUERY

```
create DATABASE DBMS3_PARINITA;
USE DBMS3_PARINITA;
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (
    SELECT AVG(emp_salary)
    FROM EMPLOYEES
);
```

OUTPUT

Result Grid Filter Rows:

emp_id	emp_name	emp_salary	dept_no
2	Priya	60000.00	20
5	Karan	70000.00	40
NULL	NULL	NULL	NULL